

Intelligent Agents

Homework #2

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April 5, 2018

Show which concession criteria for multilateral negotiation are deadlock-free and which are not. let $\mathcal{A}$ be the set of all agents, and $i \in \mathcal{A}$ is an agent. Let $\mathcal{A}_{others} \subset \mathcal{A}$ be the set of agents other than the one being currently considered.

Below are each of the seven concession criteria:

strong $\rightarrow$ the proposal is strictly better $\forall i \in \mathcal{A}_{others}$.

weak $\rightarrow \exists i \in \mathcal{A}_{others}$ for whom the proposal is strictly better.

Pareto $\rightarrow$ the proposal is no worse $\forall i \in \mathcal{A}_{others}$ and $\exists i \in \mathcal{A}_{others}$ for which it is strictly better.

utilitarian $\rightarrow$ proposal makes $\sum_{i \in \mathcal{A}_{others}} u_i$ increase.

egalitarian $\rightarrow$ proposal makes it so that $(i \in \mathcal{A}_{others})(\min(u_i))$ increases

Nash $\rightarrow$ proposal makes $\prod_{i \in \mathcal{A}_{others}} u_i$ increase.

egocentric $\rightarrow$ proposal is worse for the maker.

Want to show: Weak, utilitarian, and egocentric criteria are deadlock-free. Egalitarian, strong, and Pareto criteria are not deadlock-free. Nash criterion is deadlock-free iff the rule is that all utilities must be positive.

Claim 1: Pareto and strong criteria are not deadlock-free.

Proof of Claim 1: Because strong criteria are also Pareto criteria, if we show that Pareto criteria are not deadlock-free it then follows that strong criteria are not as well.

In a situation where multilateral negotiation is necessary (e.g. 3+ agents) and $|\mathcal{X}| = |\mathcal{A}|$ (where each agent only has one proposal), there can be a case where none of the proposals are not worse for all of and better for at least one of the others, such as below:

		Proposal		
		i_1	i_2	i_3
Utility	u_1	9	9	1
	u_2	9	9	2
	u_3	9	9	10

In this case, because these are the only possible outcomes they do not fulfill Pareto criteria and also cannot have been Pareto "concessions". Agent i_3 has made a proposal with u_3 maximum for itself, but utilities less than their own proposed ones for i_2 and i_1 .

Claim 2: Egalitarian criteria are not deadlock-free.

Proof of Claim 2: Let u_ϵ be a minimum possible utility for any agent in a given negotiation. Then for any negotiation where $\forall x \in \mathcal{X}$, if $\exists 2i \in \mathcal{A}$ such that both of their $u_i = u_\epsilon$, then it's impossible to have any proposal that increases the minimum utility amongst other agents because u_ϵ is the same value $\forall x \in \mathcal{X}$.

Claim 3: Utilitarian criteria are deadlock-free.

Proof of Claim 3: Let us consider a current round of proposals in the negotiation. Use i to refer to the agent whose proposal x_i is s.t. $\sum_{k \in \mathcal{A}} u_k(x_i)$ is maximum for the current round. We can show that if at this stage there is no agreement, then there is some agent $j \in \mathcal{A}$ who has their own proposal where $u_j(x_j) > u_j(x_i)$ (1). Due to the condition that x_i currently yields the maximum utility sum, we also know that $\sum_{k \in \mathcal{A}} u_k(x_i) \geq \sum_{k \in \mathcal{A}} u_k(x_j)$ (2). Then (1), (2) $\rightarrow \sum_{k \in \mathcal{A} \setminus \{j\}} u_k(x_i) > \sum_{k \in \mathcal{A} \setminus \{j\}} u_k(x_j)$, which implies that the current round with x_i satisfies the utilitarian criterion as a concession for j .

Claim 4: Weak concessions are deadlock-free.

Proof of Claim 4: During the negotiation, if an agreement isn't already reached, that means there must be some agent j such that $u_j(x_j)$ for some x_j of agent j is greater than u_j for all of the current proposals in the round. So that means there can still be the proposal x_j made according to the weak criteria if there is no agreement already and because this " x_j " if needed, can always be done as a next step, that breaks the deadlock possibility.

Claim 5: Egocentric concessions are deadlock-free.

Proof of Claim 5: The same argument for Claim 4 can be used to demonstrate deadlock freedom of egocentric concessions. We recall that a multilateral agreement is reached if $\exists i \in \mathcal{A}$ s.t. $u_j(x_i) \geq u_j(x_j)$, $\forall j \in \mathcal{A}$. If this is not true, then it means that $\exists j$ s.t. $u_j(x_i) < u_j(x_j)$. By definition of egocentric criteria, x_i at that stage has already been an egocentric concession with respect to x_j and we have also shown that weak concessions are deadlock free (Claim 4), so as long as a weak concession can be made and there isn't yet agreement, each weak concession without agreement is an egocentric concession with respect to the one or more agents whose utility for that proposal isn't high enough to satisfy agreement.

Claim 6: Nash concessions are deadlock-free iff the utility values are positive.

Proof of Claim 6: When comparing Nash to utilitarian criteria, they are very similar except Nash deals with products and the utilitarian criteria deals with summation. The nonzero requirement is then clear because if zero value utilities are allowed, and we have the condition listed in the proof of Claim 2 (egalitarian) where $u_e = 0$, then the minimum utility can't be guaranteed to increase without a deadlock. If this can't be guaranteed then it's clear that the Nash criteria can't be satisfied in this case since if even one zero value can't be changed then the product of the values cannot be anything other than zero, and cannot increase.

By proving the above 6 claims, the main claim has been satisfied. $\square$